

OM PRAKASH SINGH

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PROFILE SYNOPSIS

1. Led Unikey* **Client Onboarding** within the Mercedes domain, enabling **secure key creation** and **utilization** via **Unikey REST APIs** for **signing** and **encryption** operations. (*Unikey: Team name)
2. Served as **Concept Owner** for the **Unikey Key Management System (KMS UI)** — a self-service portal facilitating **project creation** and complete **key lifecycle operations** (create, import, export, delete) as well the **Users RBAC**.
3. Authored **POC** and **MVP** documentation, aligning **Security Architects** across **India** and **Germany**, and secured approval to initiate **full-scale development**.
4. Defined **Feature Specifications** and **Test Acceptance Criteria**, supporting **development** and **testing** under **Lean** and **Agile** methodologies.
5. Built **Pytest**, **GUI automation (Playwright Async)** and **FastAPI-based REST API** validation frameworks to monitor API service health and handle Bad ECU Certificate scenarios.
6. Designed and implemented a **socket-based multi-user solution** for **Harbor image** pushes and **Cloud Foundry deployments**, eliminating credential sharing and improving security.
7. Developed and maintained an **on-premises Jenkins CI/CD pipeline**, automating **end-to-end deployment** and **testing workflows**.
8. **Containerized** a **Django-based ETL backend** with **PostgreSQL** and **Grafana**, leveraging **Harbor** for image storage and **Cloud Foundry** for PaaS deployment.
9. Created **Grafana dashboards** and **Django-based UIs** to visualize metrics, **automate Sphinx documentation**, and streamline report generation.
10. Automated **integration** and **HIL testing** using Vector Canoe, Debugger and SSH APIs; centralized all test results into a **unified Grafana dashboard** for monitoring and analysis.

KEY SKILLS

Programming & Frameworks	Python, SQL, Django, FastAPI, Sphinx (doc-as-code)
Cloud & Containerization	AWS (EC2, S3), Azure DevOps, Docker, Cloud Foundry, Harbor
CI/CD & DevOps Tools	Jenkins, GitHub, GitLab, Bitbucket
Database & Visualization	MySQL, PostgreSQL, Grafana, Datadog
Test Automation & Scripting	Pytest, Playwright (Async), REST API Automation
Project Management & Methodologies	JIRA, Agile, Lean, PI Planning
Version Control & Collaboration	Git, GitHub, GitLab
IDE & OS Environment	VS Code, VMware, WSL2

CERTIFICATION

Name	Organization	Year
ISTQB Certified Tester Foundation Level 4.0v	Indian Testing Board	Oct 2025
Certified SAFe PO/PM	Scaled Agile, Inc	Apr 2025 – Apr 2026
Docker Foundation Professional Certification	LinkedIn	Apr 2024
AWS Certified Cloud Practitioner	Amazon Web Services	Sep 2023 – Sep 2026
ISO TÜV SÜD 21434 Automotive Cyber Security	TÜV SÜD	Dec 2025

PROFESSIONAL EXPERIENCE

Company: Mercedes-Benz Research and Development (MBRDI), Bengaluru / India Year: January 2024 – Present Position: Senior Engineer	
Project 2	Unikey (2024 – Present) - Led Unikey Client Onboarding support for users within the Mercedes domain to enable secure key creation and utilization via Unikey REST APIs for signing and encryption operations. - Concept Owner for the Unikey Key Management System (KMS UI) — a self-service portal enabling project creation, key lifecycle operations (create, import, export, delete) tailored to user requirements. - Authored POC and MVP documentation to align Security Architects across India and Germany; successfully secured approval to initiate development. Defined feature specifications and test acceptance criteria, supporting development under Lean and Agile principles. - Developed GUI automation (Playwright Async) and FastAPI REST API tests to monitor Unikey service health and validate Bad ECU certificate scenarios across root, intermediate, and backend components.
Project 1	Internal Dashboard (2024) - Developed a socket-based solution to enable multi-user access for pushing images to Harbor and executing Cloud Foundry (CF) PaaS deployments without sharing admin credentials, resulting in improved security and cost efficiency. - Implemented an on-premises Jenkins CI/CD pipeline to automate the end-to-end deployment process. - Containerized a Django-based ETL backend with PostgreSQL and Grafana, leveraging Harbor as the image registry and Cloud Foundry PaaS for deployment. - Designed Grafana dashboards to visualize and monitor performance metrics for cross-functional teams.
Company: Bosch Global Software Technologies (BGSW), Bengaluru / India Year: August 2019 – December 2023 Position: Senior Software Engineer	
Project 2	FORD (Automotive Compute Platform [ACP]) (2022 – 2023) - Developed and deployed Django-driven data models, database schemas, and a GUI for metric visualization; integrated Sphinx automation, Grafana dashboards, and Docker-based containerization for efficient packaging and deployment. - Partnered with the DevOps team to incorporate containerized Sphinx report automation into the CI/CD pipeline, administered Jenkins and Docker on on-premises servers, and implemented a PR bot notification system for JIRA/Git workflows (sanity-check for PR raised).
Project 1	Daimler (Integrated Driver Controller v5 [IDC5]) (2019 – 2022) - Automated Integration testing and validation using Vector Canoe, Debugger, and SSH APIs; implemented HIL tests via Jenkins pipelines and integrated all checks into a unified dashboard. - Designed a MySQL-based metrics system with Python automation, Grafana visualization, and a Django web UI for data consistency and performance monitoring.

EDUCATION

Education Qualification	College / School	Board	Duration	Percentage / CGPA
Bachelor of Engineering (ECE)	Dayananda Sagar College of Engineering, Bengaluru	VTU	2015-2019	8.97 CGPA